Sales Analysis Report

Retail company | Orders from Jan 2024 to Dec 2025

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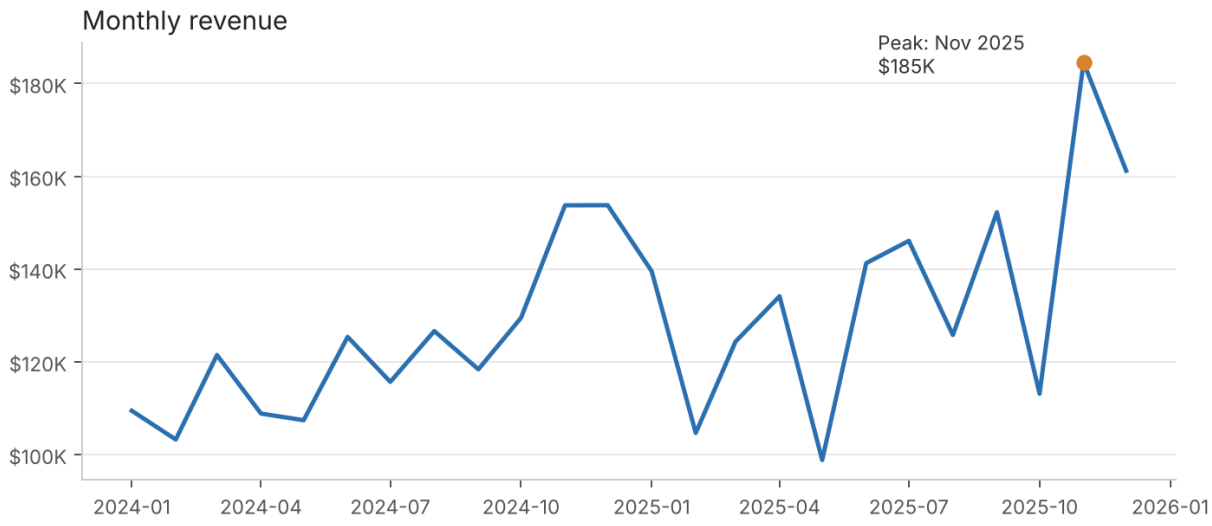
Key numbers

Metric	Value
Total revenue	$3,099,113
Total profit	$859,629
Profit margin	27.7%
Number of orders	4,383
Average order value	$707.08
Revenue growth 2025 vs 2024	+10.3%

Summary: The company earned $3,099,113 in revenue with a profit margin of 27.7%. Revenue changed by +10.3% from 2024 to 2025. **Laptop** is the best-selling product (38% of revenue). The best month was November 2025 and the weakest was May 2025.

Data note: The raw file had 4,396 rows. After removing duplicates and rows with missing quantity, 4,383 rows were used. Region names were standardised (e.g. 'north' became 'North').

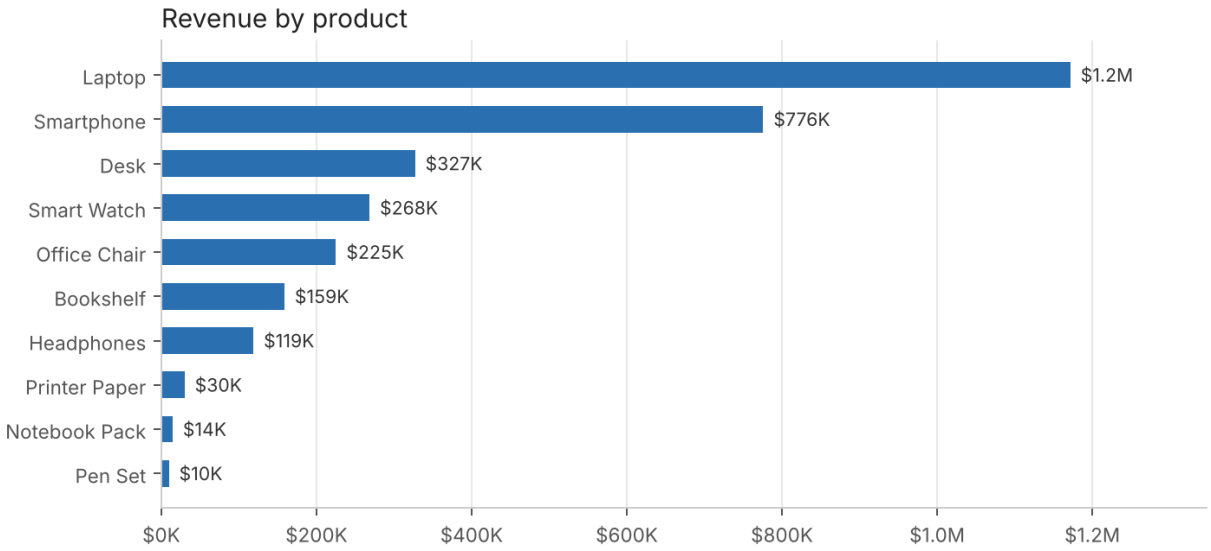
1. Sales trend



Sales show a clear seasonal pattern: average monthly revenue is clearly highest in November and December (roughly 30% above the other months), while the rest of the year is more variable. The business should plan stock and marketing before Q4.

Year	Revenue
2024	$1,473,380
2025	$1,625,733

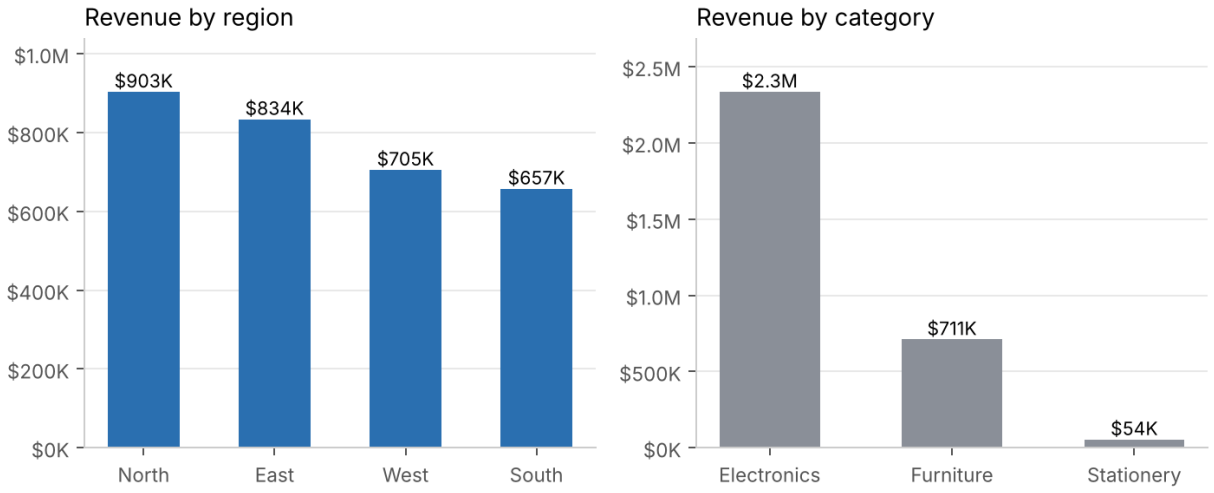
2. Products



High-priced electronics (laptops and smartphones) generate most of the revenue. Stationery sells a lot of units (about 3,800) but adds very little revenue because items are cheap, even though its profit margin percentage is the highest.

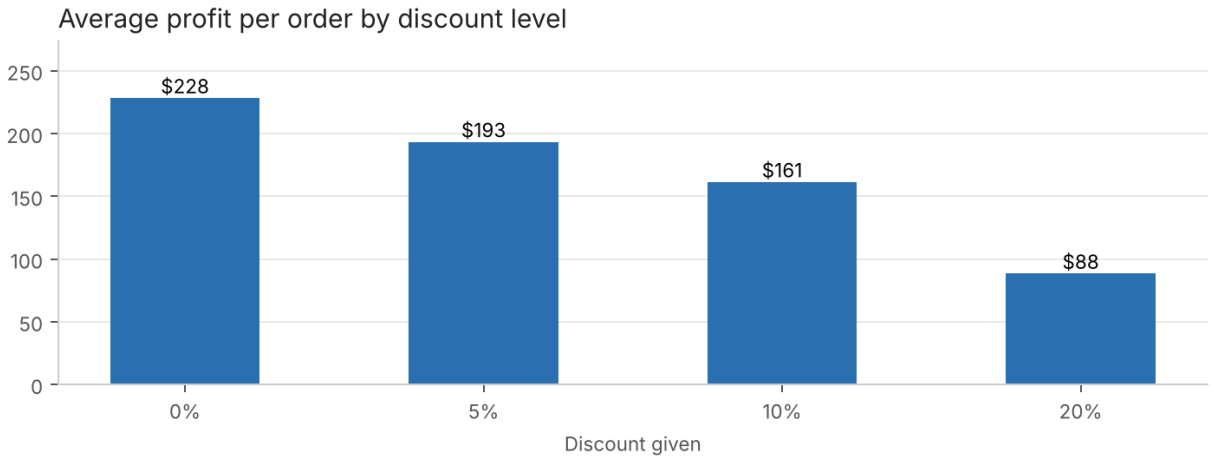
Category	Revenue	Profit	Margin
Electronics	$2,334,386	$568,658	24.4%
Furniture	$710,699	$262,427	36.9%
Stationery	$54,028	$28,544	52.8%

3. Regions and categories



North is the strongest region and **South** the weakest. The gap suggests extra marketing or a new sales rep could help the lower regions.

4. Discounts



Bigger discounts reduce profit on each order. The analysis checks whether customers buy more units to make up for it; the average quantity stays almost the same, so deep discounts are not paying for themselves.

5. Recommendations

• Stock up and run campaigns before November; that is the peak season.

• Focus on Laptop and Smartphone: they drive the most revenue.

• Limit discounts above 10%; they cut profit without raising quantity.

• Support the South region with local promotions.

• Bundle low-margin stationery with electronics to raise basket value.

Tools: Python, pandas, matplotlib, reportlab. Data is sample data created for this project.

Appendix A: Sales list

First 45 orders by date, out of 4,383 cleaned orders ($3,099,113 total revenue). The full list is created by the code as sales_data.csv.

Order	Date	Region	Product	Price	Qty	Disc.	Revenue
6083	2024-01-01	North	Bookshelf	$120	2	5%	$228.00
2488	2024-01-01	North	Desk	$260	3	20%	$624.00
3442	2024-01-01	South	Smartphone	$600	3	0%	$1,800.00
6760	2024-01-01	East	Headphones	$90	4	0%	$360.00
3316	2024-01-01	East	Laptop	$950	3	10%	$2,565.00
2996	2024-01-02	South	Office Chair	$180	2	0%	$360.00
5691	2024-01-02	West	Smart Watch	$220	5	5%	$1,045.00
6462	2024-01-02	South	Headphones	$90	4	5%	$342.00
2949	2024-01-02	North	Desk	$260	5	0%	$1,300.00
2482	2024-01-02	West	Headphones	$90	5	10%	$405.00
2409	2024-01-02	East	Desk	$260	3	10%	$702.00
2622	2024-01-03	East	Desk	$260	4	0%	$1,040.00
2913	2024-01-03	South	Headphones	$90	1	0%	$90.00
2739	2024-01-03	North	Bookshelf	$120	3	10%	$324.00
5627	2024-01-03	North	Headphones	$90	2	5%	$171.00
3646	2024-01-04	South	Bookshelf	$120	4	0%	$480.00
2550	2024-01-04	North	Notebook Pack	$12	3	0%	$36.00
1773	2024-01-04	South	Printer Paper	$25	2	5%	$47.50
2294	2024-01-04	North	Smart Watch	$220	1	5%	$209.00
4217	2024-01-04	West	Notebook Pack	$12	5	5%	$57.00
3789	2024-01-04	South	Laptop	$950	1	10%	$855.00
3628	2024-01-04	West	Pen Set	$8	4	10%	$28.80
1901	2024-01-04	South	Laptop	$950	3	0%	$2,850.00
3401	2024-01-05	South	Bookshelf	$120	5	10%	$540.00
5055	2024-01-05	West	Office Chair	$180	5	0%	$900.00
1465	2024-01-05	East	Smartphone	$600	4	10%	$2,160.00
1104	2024-01-06	West	Laptop	$950	1	0%	$950.00
5689	2024-01-06	South	Pen Set	$8	3	0%	$24.00

2279	2024-01-06	East	Desk	$260	1	5%	$247.00
4725	2024-01-07	East	Headphones	$90	1	0%	$90.00
5131	2024-01-07	West	Headphones	$90	2	0%	$180.00
5768	2024-01-07	South	Smartphone	$600	1	0%	$600.00
2175	2024-01-07	South	Smartphone	$600	3	20%	$1,440.00
2070	2024-01-08	East	Notebook Pack	$12	2	0%	$24.00
1303	2024-01-08	East	Smartphone	$600	3	10%	$1,620.00
4799	2024-01-08	North	Notebook Pack	$12	3	20%	$28.80
5788	2024-01-08	North	Office Chair	$180	5	20%	$720.00
6916	2024-01-09	West	Laptop	$950	2	0%	$1,900.00
6880	2024-01-09	East	Pen Set	$8	2	0%	$16.00
1801	2024-01-09	South	Pen Set	$8	4	5%	$30.40
4822	2024-01-09	West	Smartphone	$600	4	5%	$2,280.00
6438	2024-01-09	North	Laptop	$950	1	10%	$855.00
5393	2024-01-09	East	Bookshelf	$120	3	0%	$360.00
2682	2024-01-10	East	Bookshelf	$120	3	5%	$342.00
6078	2024-01-10	East	Smart Watch	$220	5	5%	$1,045.00

Appendix B: Python source code

```
"""
Project 1: Company Sales Analysis
---------------------------------
Steps:
  1. Create (or load) sales data
  2. Clean the data
  3. Analyze: monthly trend, top products, regions, categories, discounts
  4. Make charts
  5. Build a PDF report

Run:  python sales_analysis.py
Needs: pip install pandas numpy matplotlib reportlab
"""

import os
import numpy as np
import pandas as pd
import matplotlib
matplotlib.use("Agg")  # no window, just save images
import matplotlib.pyplot as plt

from reportlab.lib.pagesizes import A4
from reportlab.lib.styles import getSampleStyleSheet, ParagraphStyle
from reportlab.lib.units import cm
from reportlab.lib import colors
from reportlab.platypus import (SimpleDocTemplate, Paragraph, Spacer, Image,
                                Table, TableStyle, PageBreak)

BASE = os.path.dirname(os.path.abspath(__file__))
DATA_FILE = os.path.join(BASE, "sales_data.csv")
CHART_DIR = os.path.join(BASE, "charts")
PDF_FILE = os.path.join(BASE, "Sales_Analysis_Report.pdf")
os.makedirs(CHART_DIR, exist_ok=True)

BLUE = "#2a6fb0"      # main color
GREY = "#8a8f98"      # secondary
ORANGE = "#d9822b"    # highlight


# ---------------------------------------------------------------- 1. DATA
def create_sample_data(path, seed=42):
    """Make a realistic sales dataset for 2024-2025 (about 6000 orders)."""
    rng = np.random.default_rng(seed)
    products = {
        # name: (category, unit price)
        "Laptop": ("Electronics", 950), "Smartphone": ("Electronics", 600),
        "Headphones": ("Electronics", 90), "Smart Watch": ("Electronics", 220),
        "Office Chair": ("Furniture", 180), "Desk": ("Furniture", 260),
        "Bookshelf": ("Furniture", 120),
        "Notebook Pack": ("Stationery", 12), "Pen Set": ("Stationery", 8),
        "Printer Paper": ("Stationery", 25),
    }
    regions = ["North", "South", "East", "West"]
    region_weight = [0.30, 0.22, 0.26, 0.22]

    n = 6000
    dates = pd.to_datetime("2024-01-01") + pd.to_timedelta(
        rng.integers(0, 730, n), unit="D")
    names = rng.choice(list(products), n)
    df = pd.DataFrame({
        "order_id": np.arange(1001, 1001 + n),
        "order_date": dates,
        "region": rng.choice(regions, n, p=region_weight),
        "product": names,
        "category": [products[p][0] for p in names],
        "unit_price": [products[p][1] for p in names],
        "quantity": rng.integers(1, 6, n),
        "discount": rng.choice([0, 0.05, 0.10, 0.20], n, p=[0.55, 0.2, 0.15, 0.1]),
    })
    # seasonality: more sales in Nov/Dec, slow growth over time
    month = df["order_date"].dt.month
    keep_prob = np.where(month.isin([11, 12]), 1.0, 0.75)
    keep_prob = keep_prob * (0.85 + 0.15 * (df["order_date"].dt.year - 2024))
    df = df[rng.random(len(df)) < keep_prob].copy()
    df["cost_ratio"] = df["category"].map(
        {"Electronics": 0.72, "Furniture": 0.60, "Stationery": 0.45})

    # add a few dirty rows so the cleaning step is meaningful
    dirty = df.sample(25, random_state=1).index
    df.loc[dirty[:10], "region"] = df.loc[dirty[:10], "region"].str.lower()
    df.loc[dirty[10:15], "quantity"] = np.nan
    df = pd.concat([df, df.iloc[:8]])  # 8 duplicate rows
    df.to_csv(path, index=False)
```

```
def load_and_clean(path):
    df = pd.read_csv(path, parse_dates=["order_date"])
    rows_before = len(df)
    df = df.drop_duplicates()
    df["region"] = df["region"].str.strip().str.title()
    df = df.dropna(subset=["quantity"])
    df["quantity"] = df["quantity"].astype(int)

    # new columns
    df["revenue"] = df["unit_price"] * df["quantity"] * (1 - df["discount"])
    df["profit"] = df["revenue"] - df["unit_price"] * df["quantity"] * df["cost_ratio"]
    df["month"] = df["order_date"].dt.to_period("M").dt.to_timestamp()
    df["year"] = df["order_date"].dt.year
    print(f"Rows: {rows_before} -> {len(df)} after cleaning")
    return df, rows_before


# ------------------------------------------------------------ 2. ANALYSIS
def analyze(df):
    r = {}
    r["total_revenue"] = df["revenue"].sum()
    r["total_profit"] = df["profit"].sum()
    r["margin"] = r["total_profit"] / r["total_revenue"] * 100
    r["orders"] = df["order_id"].nunique()
    r["aov"] = r["total_revenue"] / r["orders"]

    r["by_year"] = df.groupby("year")["revenue"].sum()
    r["growth"] = (r["by_year"].loc[2025] / r["by_year"].loc[2024] - 1) * 100
    r["monthly"] = df.groupby("month")["revenue"].sum()
    r["product"] = df.groupby("product")["revenue"].sum().sort_values(ascending=False)
    r["category"] = df.groupby("category")[["revenue", "profit"]].sum()
    r["category"]["margin %"] = r["category"]["profit"] / r["category"]["revenue"] * 100
    r["region"] = df.groupby("region")["revenue"].sum().sort_values(ascending=False)
    r["discount"] = df.groupby("discount")[["quantity", "profit"]].agg(
        {"quantity": "mean", "profit": "mean"})
    r["best_month"] = r["monthly"].idxmax()
    r["worst_month"] = r["monthly"].idxmin()
    return r


# ------------------------------------------------------------ 3. CHARTS
def style_axes(ax):
    for s in ("top", "right"):
        ax.spines[s].set_visible(False)
    ax.spines["left"].set_color("#cccccc")
    ax.spines["bottom"].set_color("#cccccc")
    ax.tick_params(colors="#555555", labelsize=9)
    ax.grid(axis="y", color="#e8e8e8", linewidth=0.8)
    ax.set_axisbelow(True)


def money(x, _=None):
    return f"${x/1000:,.0f}K" if abs(x) < 1e6 else f"${x/1e6:,.1f}M"


def make_charts(r):
    paths = {}

    # a) monthly revenue line
    fig, ax = plt.subplots(figsize=(8, 3.6))
    m = r["monthly"]
    ax.plot(m.index, m.values, color=BLUE, linewidth=2)
    ax.scatter([r["best_month"]], [m.max()], color=ORANGE, zorder=3, s=40)
    ax.annotate(f"Peak: {r['best_month']:%b %Y}\n{money(m.max())}",
                (r["best_month"], m.max()), xytext=(-95, -5),
                textcoords="offset points", fontsize=9, color="#333333")
    ax.yaxis.set_major_formatter(plt.FuncFormatter(money))
    ax.set_title("Monthly revenue", loc="left", fontsize=12, color="#222222")
    style_axes(ax)
    fig.tight_layout()
    paths["monthly"] = os.path.join(CHART_DIR, "monthly_revenue.png")
    fig.savefig(paths["monthly"], dpi=180); plt.close(fig)

    # b) top products bar
    fig, ax = plt.subplots(figsize=(8, 3.8))
    p = r["product"].sort_values()
    ax.barh(p.index, p.values, color=BLUE, height=0.6)
    for y, v in enumerate(p.values):
        ax.text(v, y, "  " + money(v), va="center", fontsize=8.5, color="#333333")
    ax.xaxis.set_major_formatter(plt.FuncFormatter(money))
    ax.set_xlim(0, p.max() * 1.15)
    ax.set_title("Revenue by product", loc="left", fontsize=12, color="#222222")
    style_axes(ax)
    ax.grid(axis="y", visible=False)
    ax.grid(axis="x", color="#e8e8e8", linewidth=0.8)
    fig.tight_layout()
    paths["product"] = os.path.join(CHART_DIR, "product_revenue.png")
```

```
    fig.savefig(paths["product"], dpi=180); plt.close(fig)

    # c) region + category side by side
    fig, (a1, a2) = plt.subplots(1, 2, figsize=(8, 3.4))
    rg = r["region"]
    a1.bar(rg.index, rg.values, color=BLUE, width=0.55)
    for x, v in enumerate(rg.values):
        a1.text(x, v, money(v), ha="center", va="bottom", fontsize=8.5)
    a1.yaxis.set_major_formatter(plt.FuncFormatter(money))
    a1.set_title("Revenue by region", loc="left", fontsize=11)
    a1.set_ylim(0, rg.max() * 1.15)
    style_axes(a1)

    cat = r["category"]["revenue"].sort_values(ascending=False)
    a2.bar(cat.index, cat.values, color=GREY, width=0.55)
    for x, v in enumerate(cat.values):
        a2.text(x, v, money(v), ha="center", va="bottom", fontsize=8.5)
    a2.yaxis.set_major_formatter(plt.FuncFormatter(money))
    a2.set_title("Revenue by category", loc="left", fontsize=11)
    a2.set_ylim(0, cat.max() * 1.15)
    style_axes(a2)
    fig.tight_layout()
    paths["region_cat"] = os.path.join(CHART_DIR, "region_category.png")
    fig.savefig(paths["region_cat"], dpi=180); plt.close(fig)

    # d) discount vs average profit per order line
    fig, ax = plt.subplots(figsize=(8, 3.2))
    d = r["discount"]
    labels = [f"{int(x*100)}%" for x in d.index]
    ax.bar(labels, d["profit"].values, color=BLUE, width=0.5)
    for x, v in enumerate(d["profit"].values):
        ax.text(x, v, f"${v:,.0f}", ha="center", va="bottom", fontsize=9)
    ax.set_xlabel("Discount given", fontsize=9, color="#555555")
    ax.set_title("Average profit per order by discount level",
                 loc="left", fontsize=12, color="#222222")
    ax.set_ylim(0, d["profit"].max() * 1.2)
    style_axes(ax)
    fig.tight_layout()
    paths["discount"] = os.path.join(CHART_DIR, "discount_profit.png")
    fig.savefig(paths["discount"], dpi=180); plt.close(fig)
    return paths


# ---------------------------------------------------------------- 4. PDF
def build_pdf(df, r, charts, rows_before):
    ss = getSampleStyleSheet()
    H1 = ParagraphStyle("H1", parent=ss["Heading1"], textColor=colors.HexColor("#1d4f80"))
    H2 = ParagraphStyle("H2", parent=ss["Heading2"], textColor=colors.HexColor("#1d4f80"))
    body = ParagraphStyle("body", parent=ss["Normal"], fontSize=10.5, leading=15)
    doc = SimpleDocTemplate(PDF_FILE, pagesize=A4, leftMargin=2*cm,
                            rightMargin=2*cm, topMargin=2*cm, bottomMargin=2*cm,
                            title="Sales Analysis Report")
    S = []

    def table(data, widths=None):
        t = Table(data, colWidths=widths)
        t.setStyle(TableStyle([
            ("BACKGROUND", (0, 0), (-1, 0), colors.HexColor("#2a6fb0")),
            ("TEXTCOLOR", (0, 0), (-1, 0), colors.white),
            ("FONTSIZE", (0, 0), (-1, -1), 9.5),
            ("ROWBACKGROUNDS", (0, 1), (-1, -1), [colors.white, colors.HexColor("#f2f5f9")]),
            ("GRID", (0, 0), (-1, -1), 0.4, colors.HexColor("#cccccc")),
            ("ALIGN", (1, 0), (-1, -1), "RIGHT"),
            ("TOPPADDING", (0, 0), (-1, -1), 5), ("BOTTOMPADDING", (0, 0), (-1, -1), 5),
        ]))
        return t

    # Title
    S += [Spacer(1, 3*cm), Paragraph("Sales Analysis Report", ss["Title"]),
          Paragraph("Retail company | Orders from Jan 2024 to Dec 2025",
                    ParagraphStyle("c", parent=body, alignment=1)),
          Spacer(1, 0.4*cm),
          Paragraph("<b>Student name:</b> Syed Hamza Ali", ParagraphStyle("nm", parent=body, alignment=1, fontSize=12)),
          Spacer(1, 1*cm)]

    # Key numbers
    kpi = [["Metric", "Value"],
           ["Total revenue", f"${r['total_revenue']:,.0f}"],
           ["Total profit", f"${r['total_profit']:,.0f}"],
           ["Profit margin", f"{r['margin']:.1f}%"],
           ["Number of orders", f"{r['orders']:,}"],
           ["Average order value", f"${r['aov']:,.2f}"],
           ["Revenue growth 2025 vs 2024", f"{r['growth']:+.1f}%"]]
    S += [Paragraph("Key numbers", H2), table(kpi, [9*cm, 5*cm]), Spacer(1, 0.6*cm)]

    top = r["product"].index[0]
    top_share = r["product"].iloc[0] / r["total_revenue"] * 100
```

```
S.append(Paragraph(
    f"<b>Summary:</b> The company earned ${r['total_revenue']:,.0f} in revenue "
    f"with a profit margin of {r['margin']:.1f}%. Revenue changed by "
    f"{r['growth']:+.1f}% from 2024 to 2025. <b>{top}</b> is the best-selling "
    f"product ({top_share:.0f}% of revenue). The best month was "
    f"{r['best_month']:%B %Y} and the weakest was {r['worst_month']:%B %Y}.", body))
S.append(Spacer(1, 0.4*cm))
S.append(Paragraph(
    f"<b>Data note:</b> The raw file had {rows_before:,} rows. After removing "
    f"duplicates and rows with missing quantity, {len(df):,} rows were used. "
    "Region names were standardised (e.g. 'north' became 'North').", body))
S.append(PageBreak())

# Trend
S += [Paragraph("1. Sales trend", H1), Image(charts["monthly"], 16.5*cm, 7.4*cm),
      Spacer(1, 0.3*cm),
      Paragraph("Sales show a clear seasonal pattern: average monthly revenue "
                "is clearly highest in November and December (roughly 30% above "
                "the other months), while the rest of the year is more variable. "
                "The business should plan stock and marketing before Q4.", body),
      Spacer(1, 0.5*cm)]
yr = [["Year", "Revenue"]] + [[str(y), f"${v:,.0f}"] for y, v in r["by_year"].items()]
S += [table(yr, [5*cm, 5*cm])]
S.append(PageBreak())

# Products
S += [Paragraph("2. Products", H1), Image(charts["product"], 16.5*cm, 7.8*cm),
      Spacer(1, 0.3*cm),
      Paragraph("High-priced electronics (laptops and smartphones) generate most "
                "of the revenue. Stationery sells a lot of units (about 3,800) "
                "but adds very little revenue because items are cheap, even "
                "though its profit margin percentage is the highest.", body),
      Spacer(1, 0.4*cm)]
cat = r["category"].sort_values("revenue", ascending=False)
ct = [["Category", "Revenue", "Profit", "Margin"]] + [
    [c, f"${row.revenue:,.0f}", f"${row.profit:,.0f}", f"{row['margin %']:.1f}%"]
    for c, row in cat.iterrows()]
S += [table(ct, [4.5*cm, 4*cm, 4*cm, 3*cm])]
S.append(PageBreak())

# Region + category
S += [Paragraph("3. Regions and categories", H1),
      Image(charts["region_cat"], 16.5*cm, 7*cm), Spacer(1, 0.3*cm),
      Paragraph(f"<b>{r['region'].index[0]}</b> is the strongest region and "
                f"<b>{r['region'].index[-1]}</b> the weakest. The gap suggests "
                "extra marketing or a new sales rep could help the lower regions.",
                body),
      Spacer(1, 0.8*cm),
      Paragraph("4. Discounts", H1), Image(charts["discount"], 16.5*cm, 6.6*cm),
      Spacer(1, 0.3*cm),
      Paragraph("Bigger discounts reduce profit on each order. The analysis "
                "checks whether customers buy more units to make up for it; "
                "the average quantity stays almost the same, so deep discounts "
                "are not paying for themselves.", body)]
S.append(PageBreak())

# Recommendations
S += [Paragraph("5. Recommendations", H1)]
recs = [
    "Stock up and run campaigns before November; that is the peak season.",
    f"Focus on {r['product'].index[0]} and {r['product'].index[1]}: they drive the most revenue.",
    "Limit discounts above 10%; they cut profit without raising quantity.",
    f"Support the {r['region'].index[-1]} region with local promotions.",
    "Bundle low-margin stationery with electronics to raise basket value.",
]
for t in recs:
    S.append(Paragraph("• " + t, body))
    S.append(Spacer(1, 0.2*cm))
S.append(Spacer(1, 0.6*cm))
S.append(Paragraph("<i>Tools: Python, pandas, matplotlib, reportlab. "
                   "Data is sample data created for this project.</i>", body))
# sales list (first 45 orders)
cols = ["order_id", "order_date", "region", "product", "unit_price", "quantity", "discount", "revenue"]
head = df.sort_values("order_date").head(45)
SALES_ROWS = [["Order", "Date", "Region", "Product", "Price", "Qty", "Disc.", "Revenue"]]
for _, x in head.iterrows():
    SALES_ROWS.append([str(x.order_id), f"{x.order_date:%Y-%m-%d}", x.region, x["product"],
                       f"${x.unit_price:,.0f}", str(x.quantity), f"{x.discount:.0%}", f"${x.revenue:,.2f}"])
SALES_WIDTHS = [1.6*cm, 2.3*cm, 1.8*cm, 3.2*cm, 1.9*cm, 1.2*cm, 1.5*cm, 2.6*cm]
SALES_NOTE = (f"First 45 orders by date, out of {len(df):,} cleaned orders "
              f"(${df['revenue'].sum():,.0f} total revenue). The full list is created "
              "by the code as sales_data.csv.")

# ---------- Appendix A: sales list ----------
S.append(PageBreak())
S.append(Paragraph("Appendix A: Sales list", H1))
S.append(Paragraph(SALES_NOTE, body))
```

```
    S.append(Spacer(1, 0.3*cm))
    S.append(table(SALES_ROWS, SALES_WIDTHS))

    # ---------- Appendix B: source code ----------
    S.append(PageBreak())
    S.append(Paragraph("Appendix B: Python source code", H1))
    code_style = ParagraphStyle("code", fontName="Courier", fontSize=6.6, leading=8.2)
    with open(os.path.abspath(__file__), encoding="utf-8") as f:
        code = f.read().replace("\t", "    ").rstrip()
    from reportlab.platypus import Preformatted
    S.append(Preformatted(code, code_style))

    doc.build(S)


# ---------------------------------------------------------------- MAIN
if __name__ == "__main__":
    if not os.path.exists(DATA_FILE):
        create_sample_data(DATA_FILE)
    df, rows_before = load_and_clean(DATA_FILE)
    results = analyze(df)
    charts = make_charts(results)
    build_pdf(df, results, charts, rows_before)
    print("Done ->", PDF_FILE)
```